MAE 263F Project Proposal: DiSMech Newton

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***Abstract*— We propose DiSMech Newton, a discrete differential geometry (DDG)-based solver for rods and shells built within the GPU-native Newton physics ecosystem. We will validate DiSMech Newton and Newton's existing Vertex Block Descent (VBD) solver against analytical theory and experimentally derived laws, compare both solvers qualitatively on Newton's existing examples, and demonstrate soft–fluid coupling with DiSMech Newton.**

I. Introduction

Simulation based on discrete differential geometry (DDG) has been highly successful for deformable linear objects [1, 2] and thin shells [3], and modern implementations [4, 5, 6, 7, 8] have added modular external forces, actuation, and differentiability. However, these implementations have largely been developed in isolation, with minimal coupling to robotics simulators [9, 10] or to solvers for other physical domains such as the material point method (MPM) and smoothed particle hydrodynamics (SPH) [11]. As a result, many DDG-based robotics contributions [12, 13] rely on ad hoc implementations tailored to their specific needs, which has limited the adoption of DDG simulators in mainstream robotics research.

Newton [14], developed in collaboration with many industry partners, emphasizes a GPU-native, multiphysics-coupled ecosystem. This makes it an opportune time to build a general-purpose DDG solver that the broader community can use to address shortcomings in existing solvers. Deformable bodies in such frameworks are typically simulated with extended position-based dynamics (XPBD) [15] or Vertex Block Descent (VBD) [16]. Both methods minimize global solves in favor of local parallelism for GPU performance, which works well for volumetric deformable objects. However, because these methods update vertices locally, long-range forces in slender rods and thin shells must propagate across many vertices. Such problems therefore require many iterations to converge, and unconverged solutions exhibit unphysical softening.

We propose to develop DiSMech Newton, a DDG-based solver within the Newton ecosystem. We focus on fair comparisons with existing Newton solvers in order to recommend when each should be used, and we explore how DDG-based solvers can be coupled to other physical domains. We will accomplish this in three phases:

1) **Comparison with theory.** Validate DiSMech Newton and VBD against Euler–Bernoulli theory for rod and shell cantilever beams, and evaluate self-contact and friction against experimentally derived laws for overhand knots [17] and plectonemes [18].
2) **Qualitative benchmarks.** Compare DiSMech Newton and VBD on Newton's provided cable, cloth, and multiphysics examples.
3) **Fluid coupling.** Demonstrate two-way coupling between DiSMech Newton and SPH, and compare the results against low-Reynolds-number resistive force theory (RFT) [19].

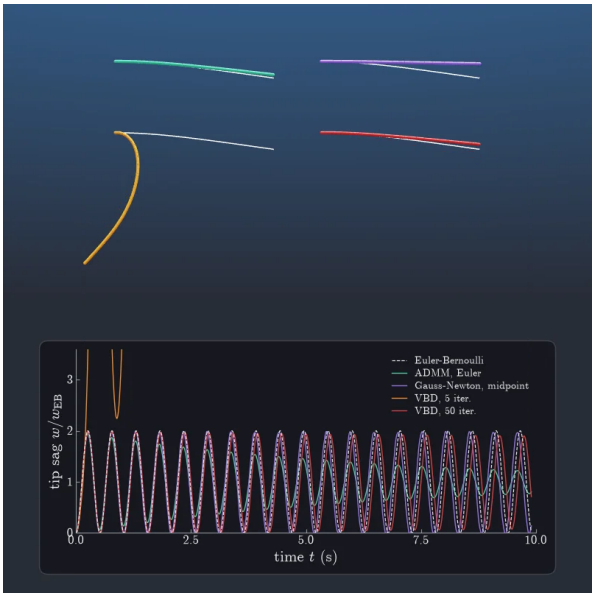


Fig. 1. Cantilever rod under gravity. DiSMech Newton with ADMM (green) and Gauss–Newton (purple) against Newton's VBD with 5 (orange) and 50 (red) iterations; white lines are the Euler–Bernoulli deflection. Under-iterated VBD softens and diverges.

II. Model

We model rods with discrete elastic rods (DER) [1, 2] and shells with midedge discrete shells [3, 20], as in DiSMech [5] (Fig. 2).

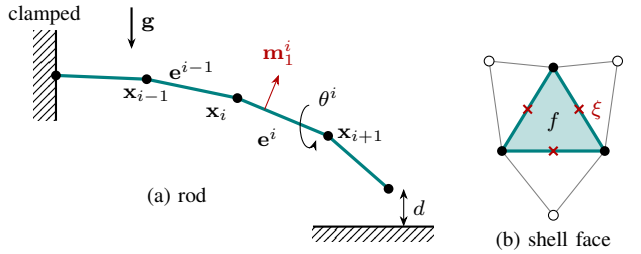
Degrees of freedom. A rod with N nodes has DOFs $\mathbf{q} = [\mathbf{x}_0, \theta^0, \mathbf{x}_1, \ldots, \mathbf{x}_{N-1}]$, $4N-1$ in total. A shell has three DOFs per vertex and one midedge angle per edge. All bodies share one global vector $\mathbf{q}$ with lumped mass matrix $\mathbf{M}$.

Energies. The elastic energy is a sum of quadratic penalties on discrete strains $\boldsymbol{\epsilon}(\mathbf{q})$ about their rest values,

$$E_{\text{el}}(\mathbf{q}) = \sum_e \tfrac{1}{2}\big(\boldsymbol{\epsilon}_e(\mathbf{q}) - \bar{\boldsymbol{\epsilon}}_e\big)^\top \mathbf{K}_e \big(\boldsymbol{\epsilon}_e(\mathbf{q}) - \bar{\boldsymbol{\epsilon}}_e\big), \qquad (1)$$

where a rod element (two consecutive edges) has stretching, bending, and twisting strains, and a shell face has membrane

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(a) rod (b) shell face

Fig. 2. Degrees of freedom. (a) A discrete elastic rod: node positions $\mathbf{x}_i$ and edge twist angles θ^i about the material frame. (b) A midedge shell face f: its vertices, the flap vertices across its edges (open), and an angle ξ per edge. Contact acts on gaps d.

Algorithm 1 DiSMech Newton time step

Require: $\mathbf{q}_n$, $\mathbf{v}_n$, h, iterations K

$\tilde{\mathbf{q}} \leftarrow \mathbf{q}_n + h\mathbf{v}_n$; $\mathbf{q} \leftarrow \tilde{\mathbf{q}}$ ▷ predict
Detect contact pairs; assemble and factor $\mathbf{A}$
for $k = 1, \ldots, K$ **do**
  for all elements e **in parallel do** ▷ local
    $\mathbf{z}_e \leftarrow \mathrm{prox}_{E_e}\left(\boldsymbol{\epsilon}_e(\mathbf{q}) + \mathbf{u}_e\right)$
    $\mathbf{u}_e \leftarrow \mathbf{u}_e + \boldsymbol{\epsilon}_e(\mathbf{q}) - \mathbf{z}_e$
  end for
  $\Delta\mathbf{q} \leftarrow -\mathbf{A}^{-1}\nabla\Phi(\mathbf{q})$ ▷ global
  $\alpha \leftarrow$ line search, capped before first impact
  $\mathbf{q} \leftarrow \mathbf{q} + \alpha\Delta\mathbf{q}$
  if a node moved past half its radius **then**
    Re-detect contact pairs
  end if
  if converged **then break**
  end if
  Update friction and contact multipliers
end for
$\mathbf{q}_{n+1} \leftarrow \mathbf{q}$; $\mathbf{v}_{n+1} \leftarrow (\mathbf{q}_{n+1} - \mathbf{q}_n)/h$

(first fundamental form) and bending (second fundamental form) strains.

Forces and boundary conditions. External forces $\mathbf{F}$ include gravity and, for fluid coupling, drag. Contact and Coulomb friction with rigid shapes, other bodies, and self-contact enter as a barrier energy $E_c(\mathbf{q})$ on the gap d of each contact pair. Clamped boundaries fix a subset of DOFs to prescribed values.

III. Algorithm

Each time step of size h solves implicit Euler as a minimization,

$$\mathbf{q}_{n+1} = \arg\min_{\mathbf{q}} \tfrac{1}{2h^2}\|\mathbf{q}-\tilde{\mathbf{q}}\|^2_{\mathbf{M}} + E_{\mathrm{el}}(\mathbf{q}) + E_c(\mathbf{q}) - \mathbf{F}^\top\mathbf{q}, \quad (2)$$

with prediction $\tilde{\mathbf{q}} = \mathbf{q}_n + h\mathbf{v}_n$. Following [21], we split E_{el} in strain space with ADMM: each element projects its strains locally in closed form, and the global step solves a linear system whose matrix $\mathbf{A}$ changes only with contact. Unlike VBD, every iteration is a global solve, so forces propagate along the whole rod in one step (Algorithm 1).

Here Φ is the objective of (2) with E_{el} replaced by its ADMM penalty $\sum_e \frac{1}{2}\|\boldsymbol{\epsilon}_e(\mathbf{q}) - \mathbf{z}_e + \mathbf{u}_e\|^2_{\mathbf{P}_e}$. For validation we also run an exact Gauss–Newton variant that refactors $\mathbf{A}$ at every iterate.

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APPENDIX